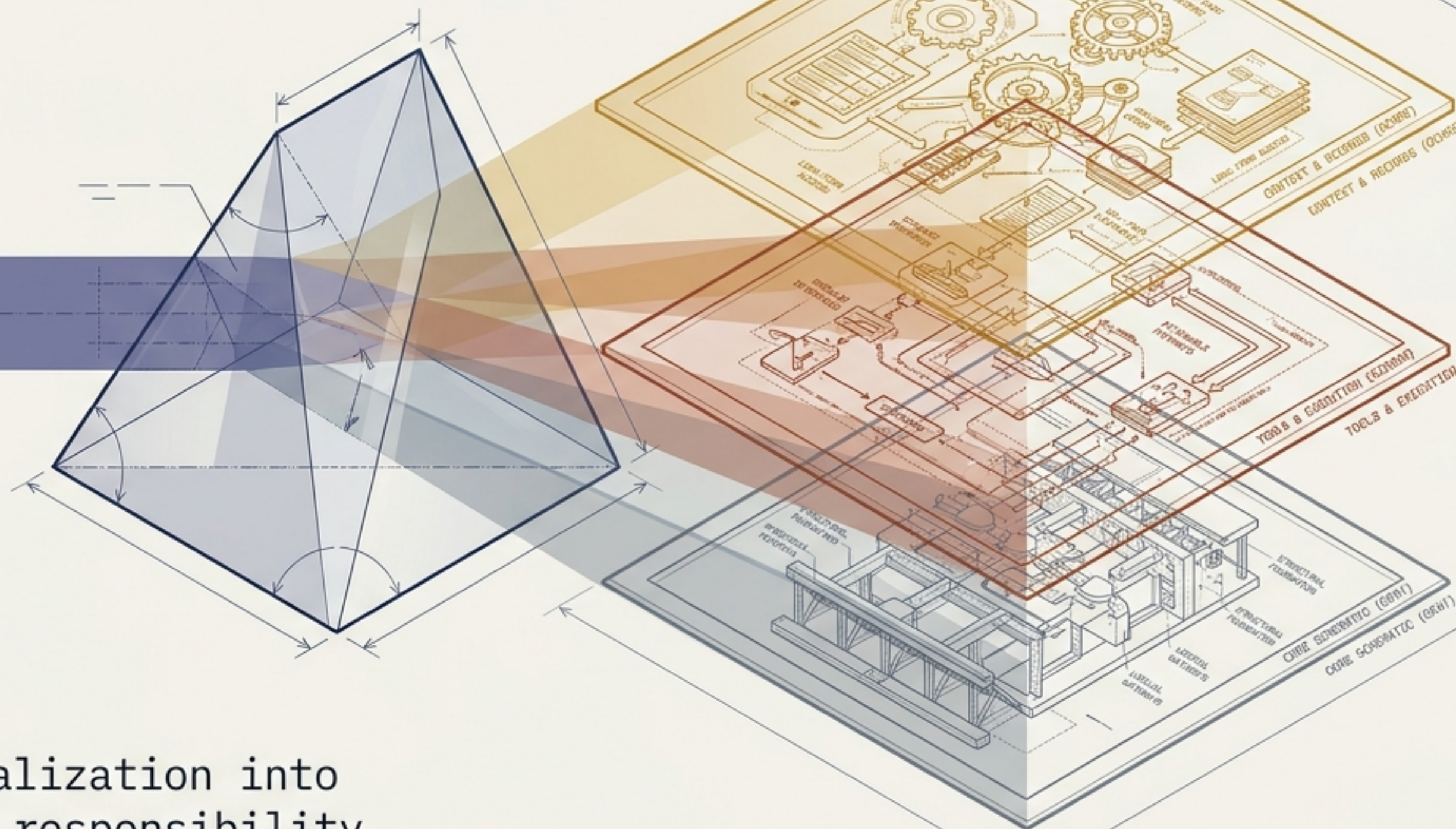


# Beyond Domain Generalization

## Relocating Specialization in AGI Architecture

GENERAL INTELLIGENCE (INDIGO)

How AGI moves specialization into context, tools, and responsibility.





# **With AGI, we may not need domain-specific agents.**

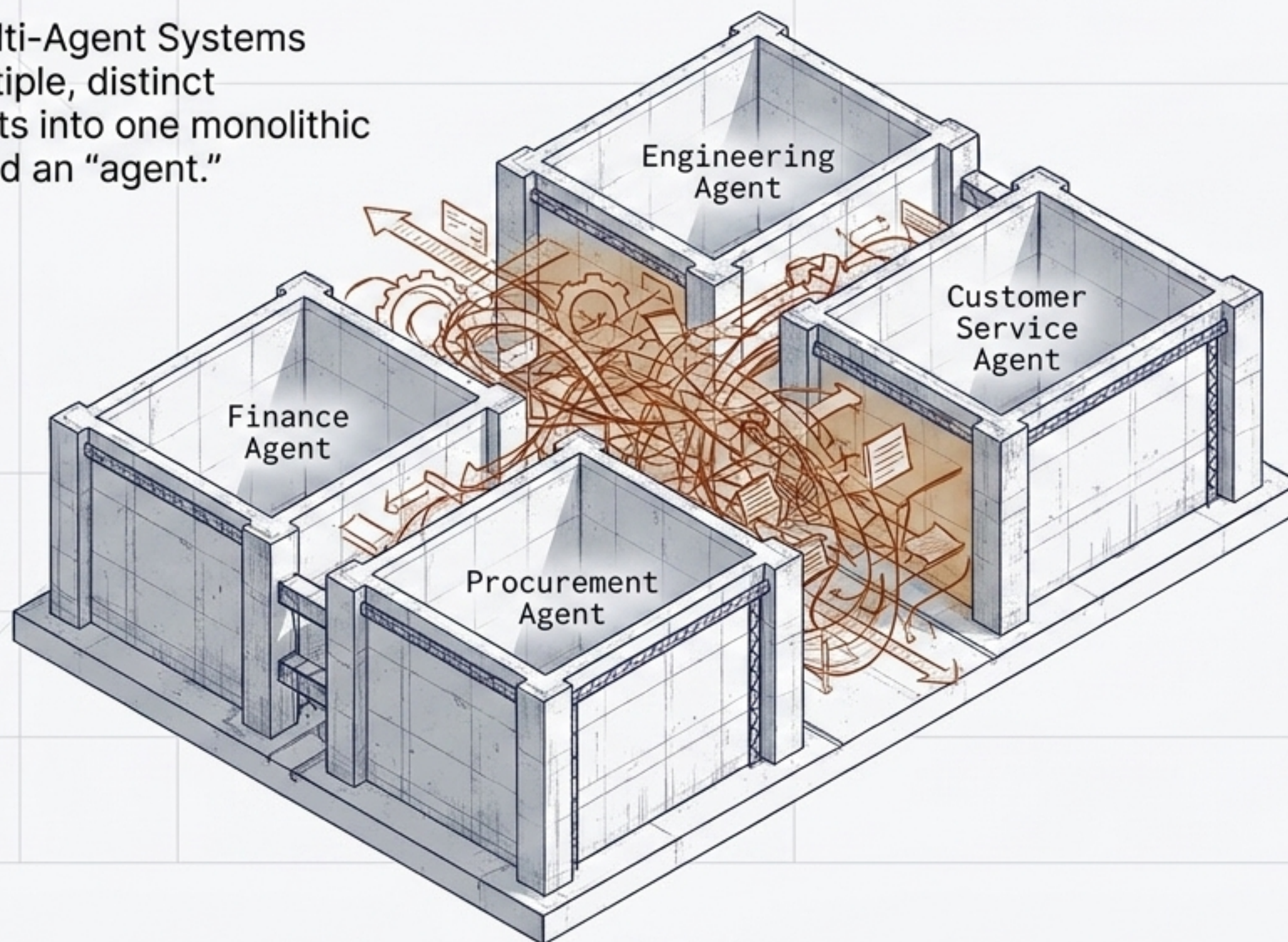
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Taken seriously, this single insight changes how every intelligent organization must be designed.



# The flaw in the current blueprint: confusing intelligence with agency

Current Multi-Agent Systems  
bundle multiple, distinct  
requirements into one monolithic  
object called an “agent.”



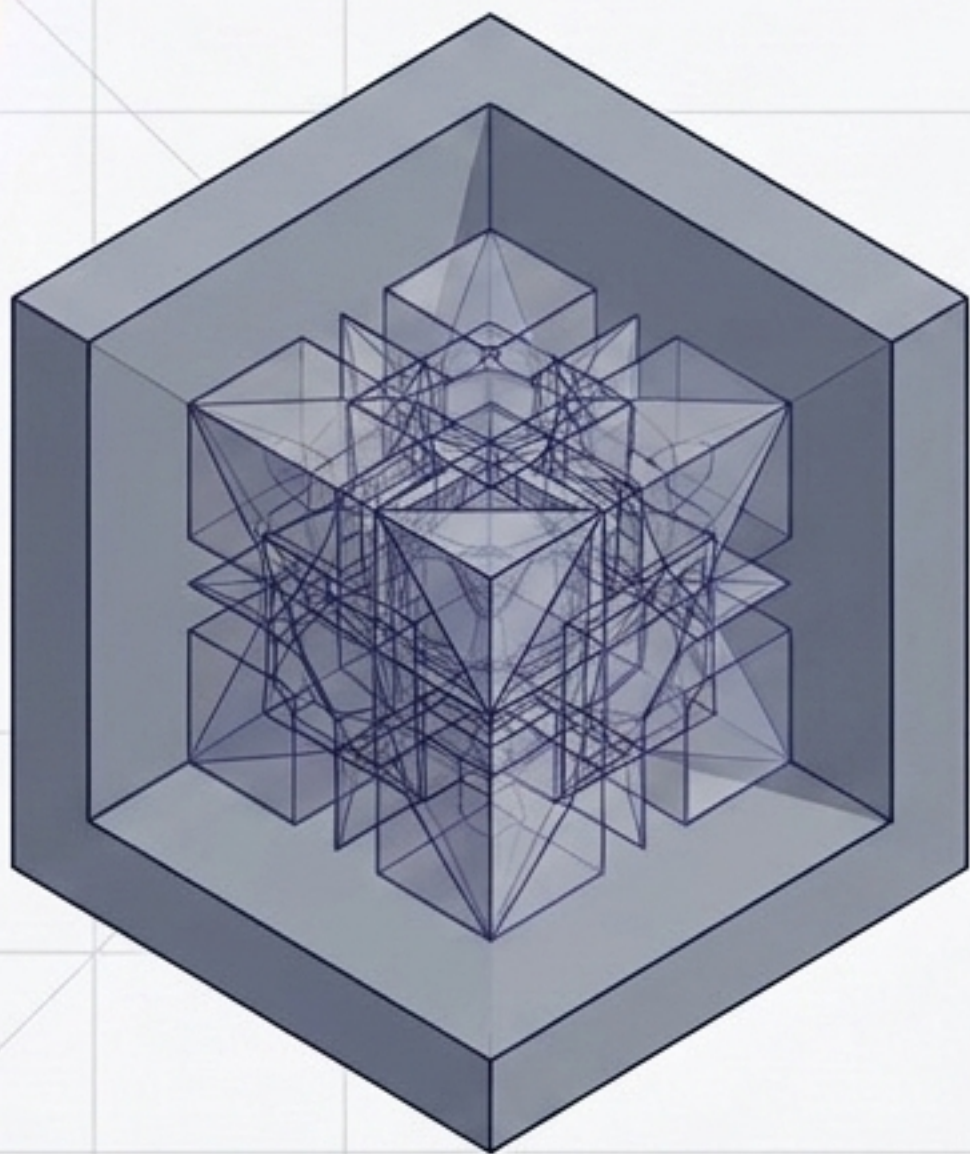
## The Monolithic Bundle

Currently, an agent is treated as a single boundary defining:

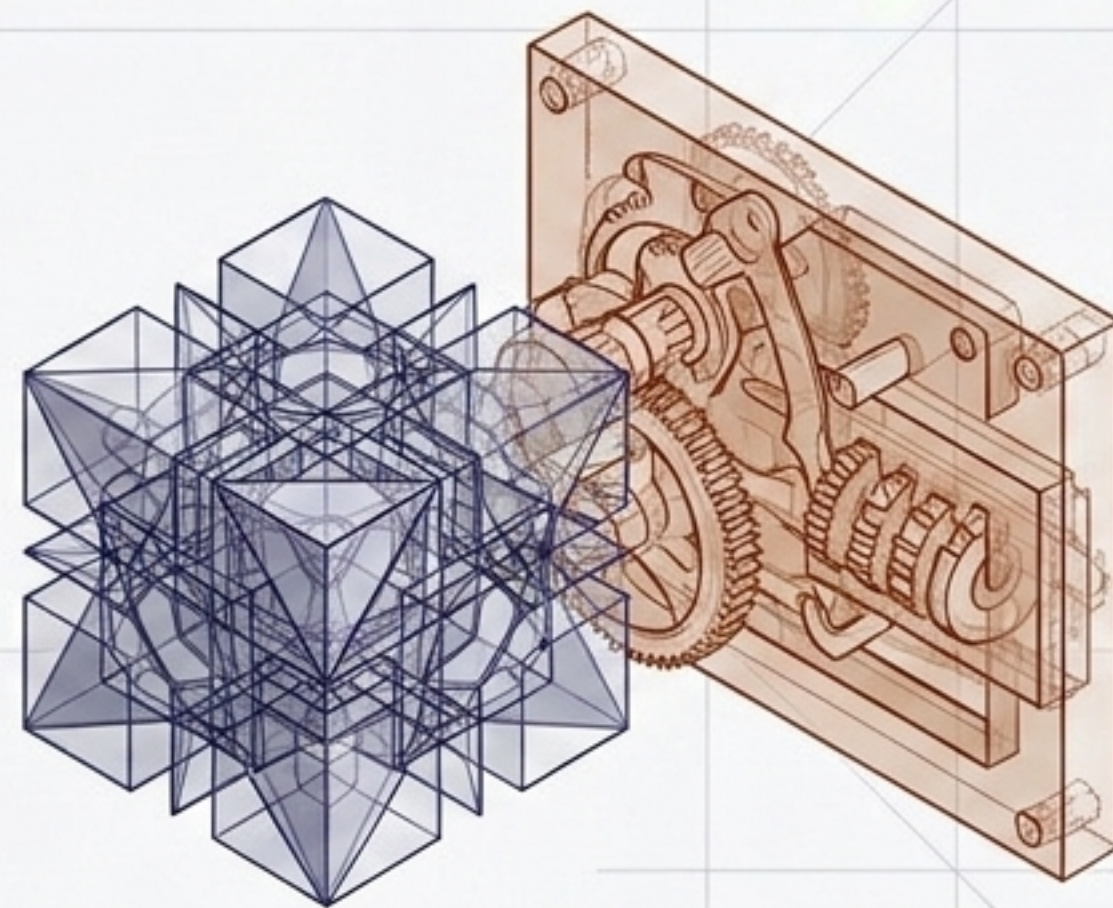
- Different knowledge
- Different software/tools
- Different human principals
- Different permissions
- Limited underlying cognitive models



Which boundaries exist because our AI is limited, and which exist because the work actually requires them?



Limits of the Intelligence

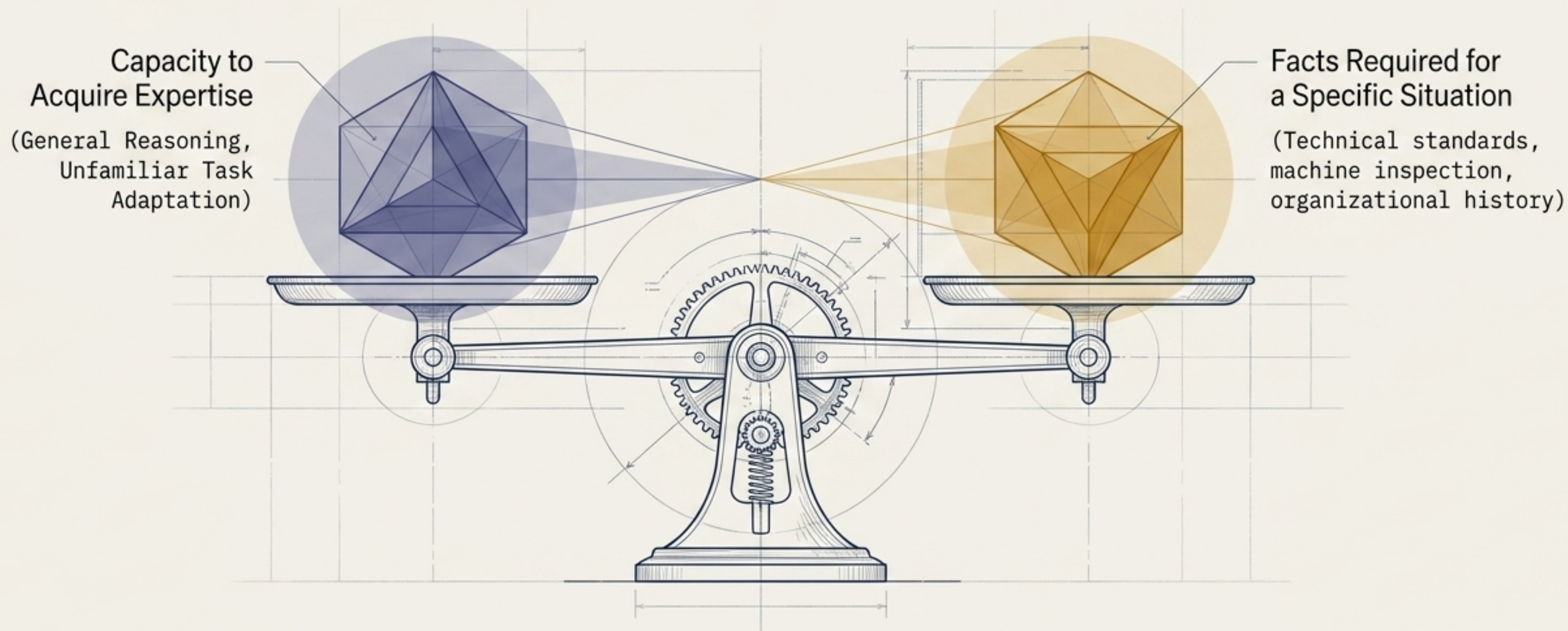


Requirements of the Work



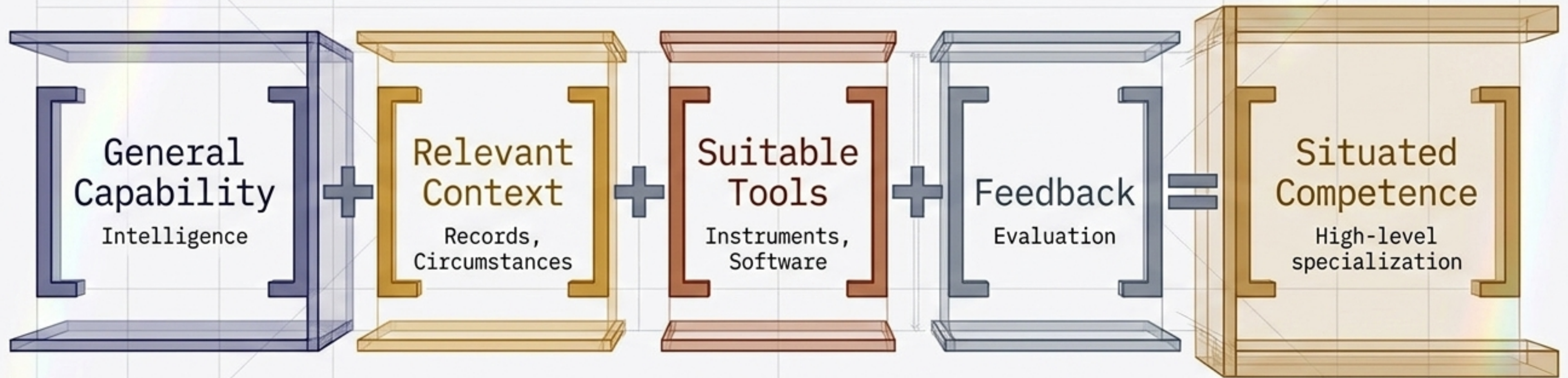
# General capability does not equal universal knowledge

AGI is a system able to acquire and apply competence across a broad range of intellectual tasks. A system could have strong general reasoning and still lack the facts needed to answer a project-specific question. That gap does not require a new agent—it requires access to records.





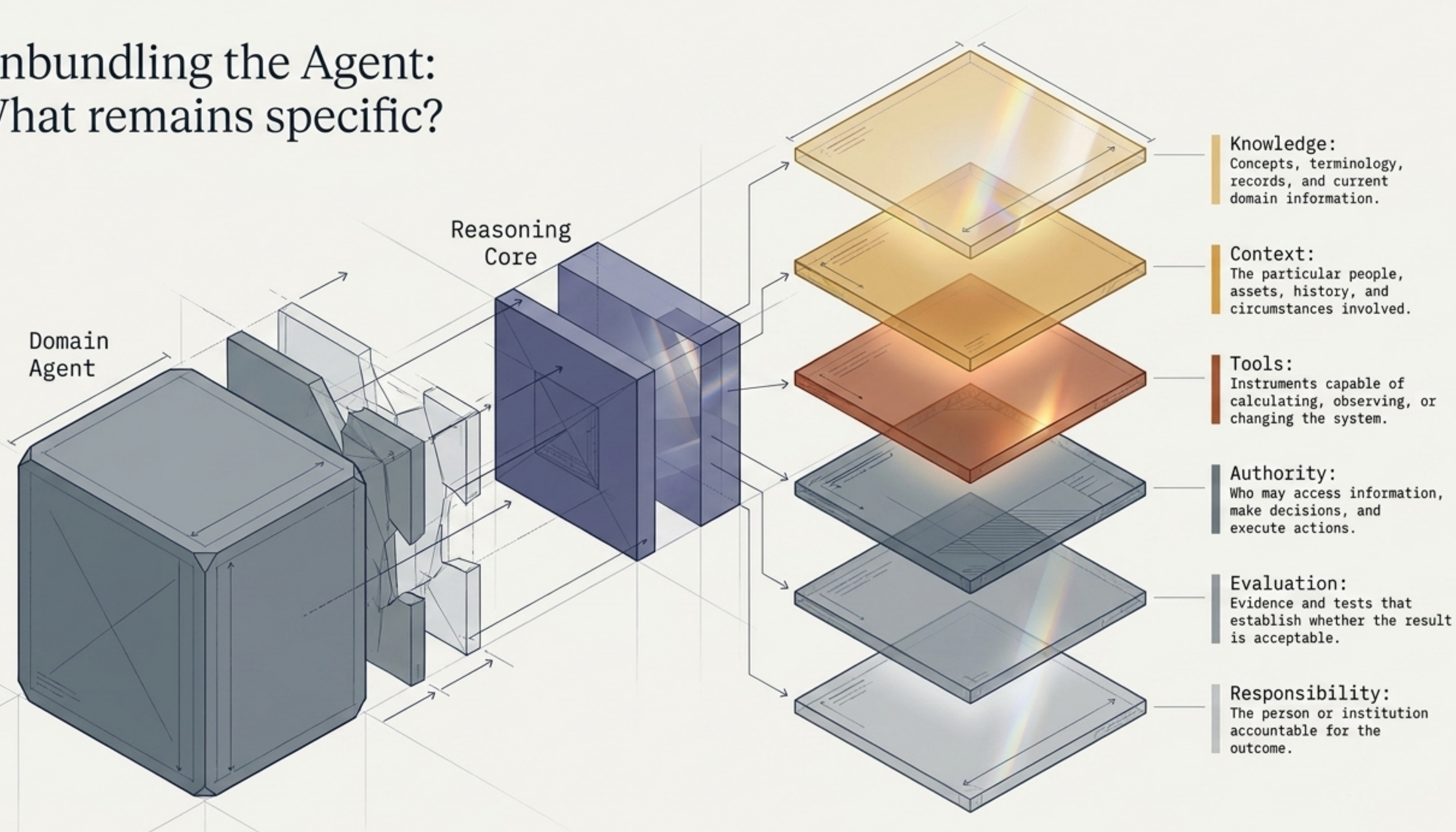
# The Equation of Situated Competence



Expertise remains necessary, but it need not always be embedded in a separately specialized intelligence.



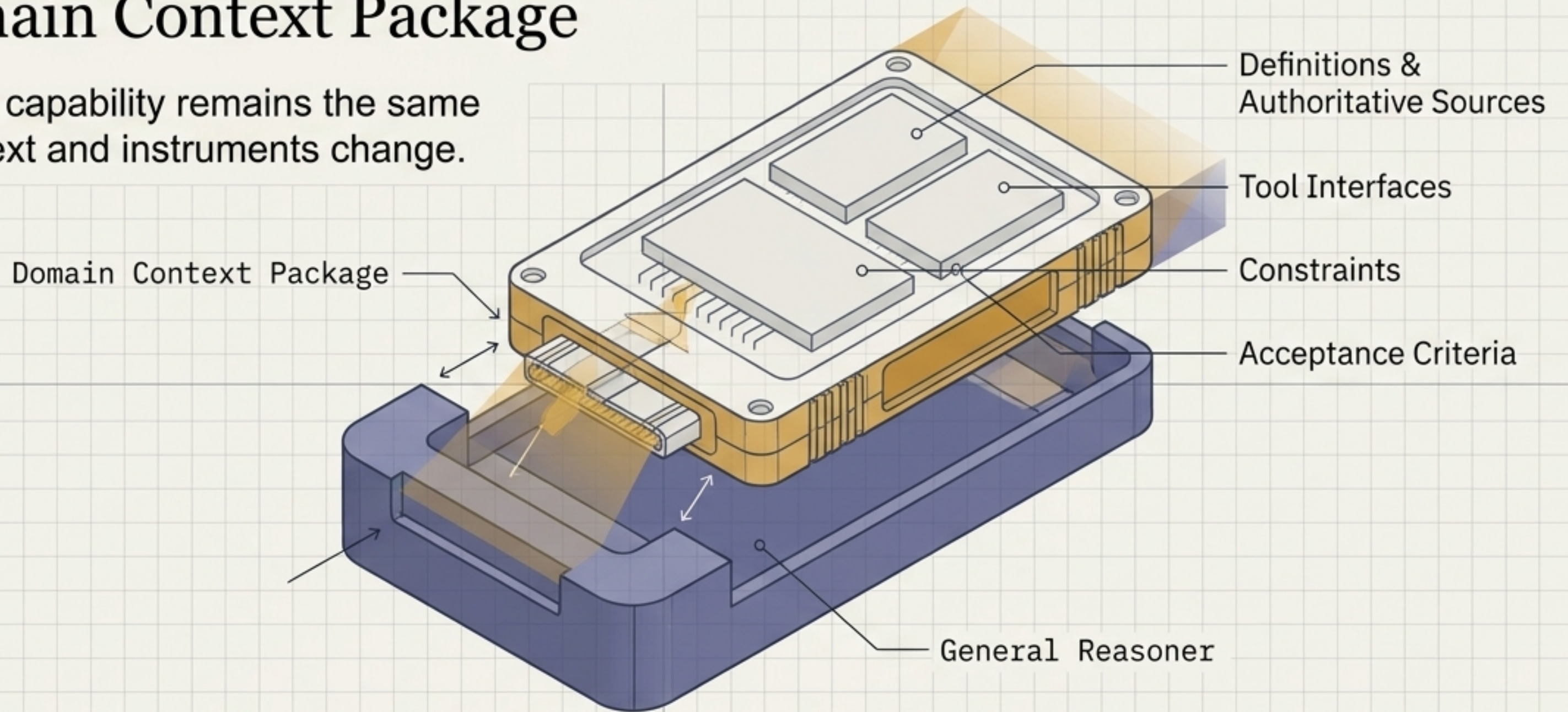
# Unbundling the Agent: What remains specific?





# The new unit of software design: The Domain Context Package

The reasoning capability remains the same while the context and instruments change.



This supports specialized work without creating an autonomous actor with its own identity, memory, and coordination overhead.

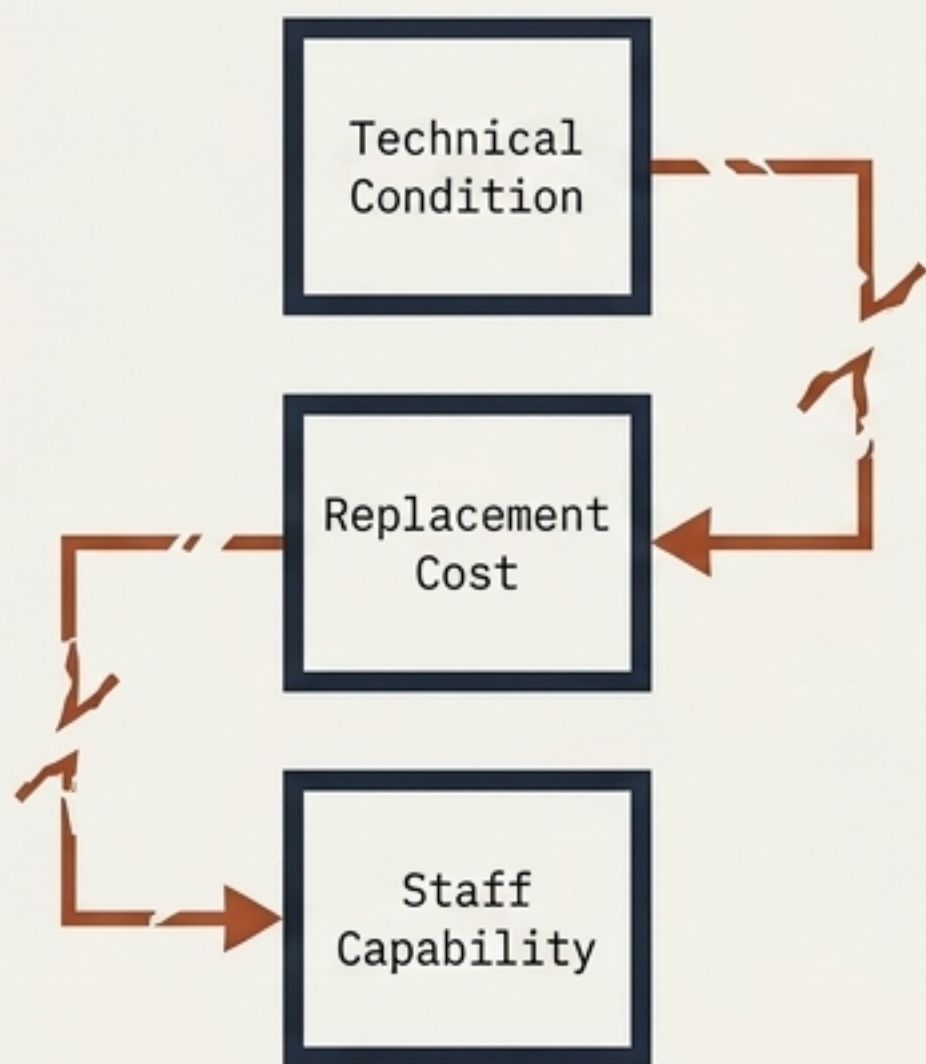


# The architectural opportunity: Cross-domain reasoning without transaction friction

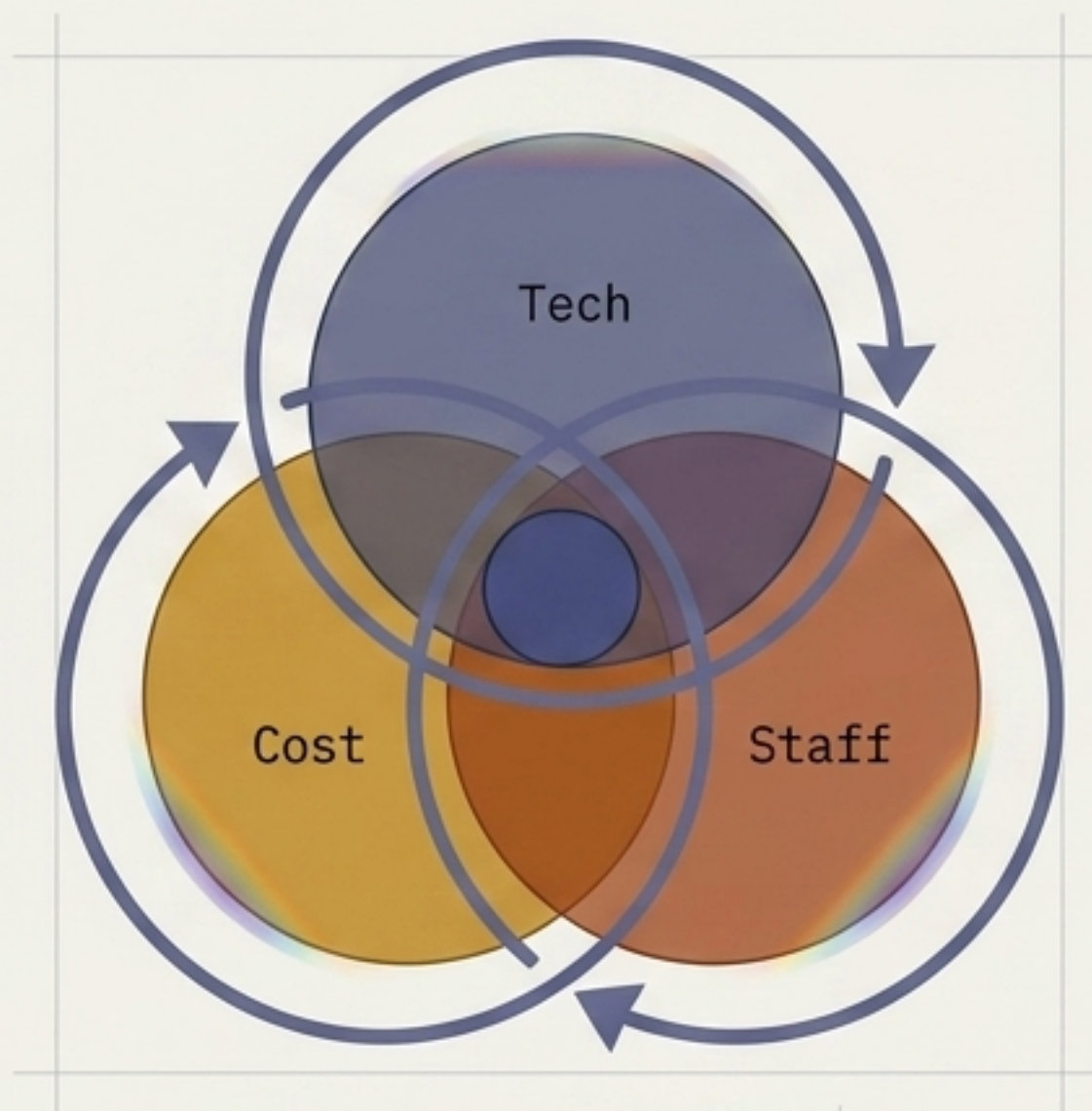
Imagine replacing unreliable equipment. A cheaper replacement requires more training.

A repair preserves cash but increases interruption risk.

## Handoff Degradation



## Coherent Representation

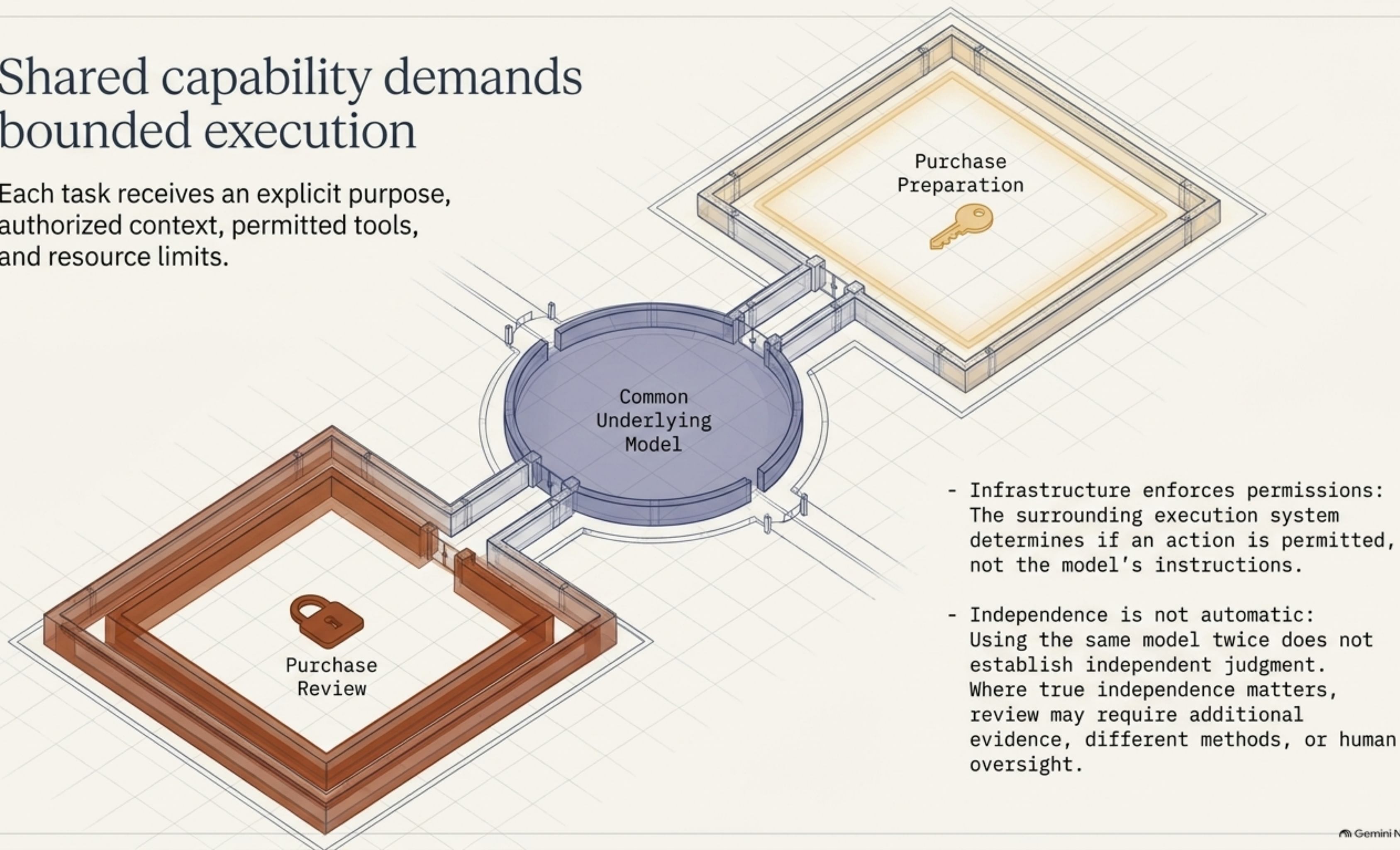


A general agent can revise a technical proposal when a staffing constraint emerges, then reconsider cost—without treating each revision as a new departmental transaction.



# Shared capability demands bounded execution

Each task receives an explicit purpose, authorized context, permitted tools, and resource limits.

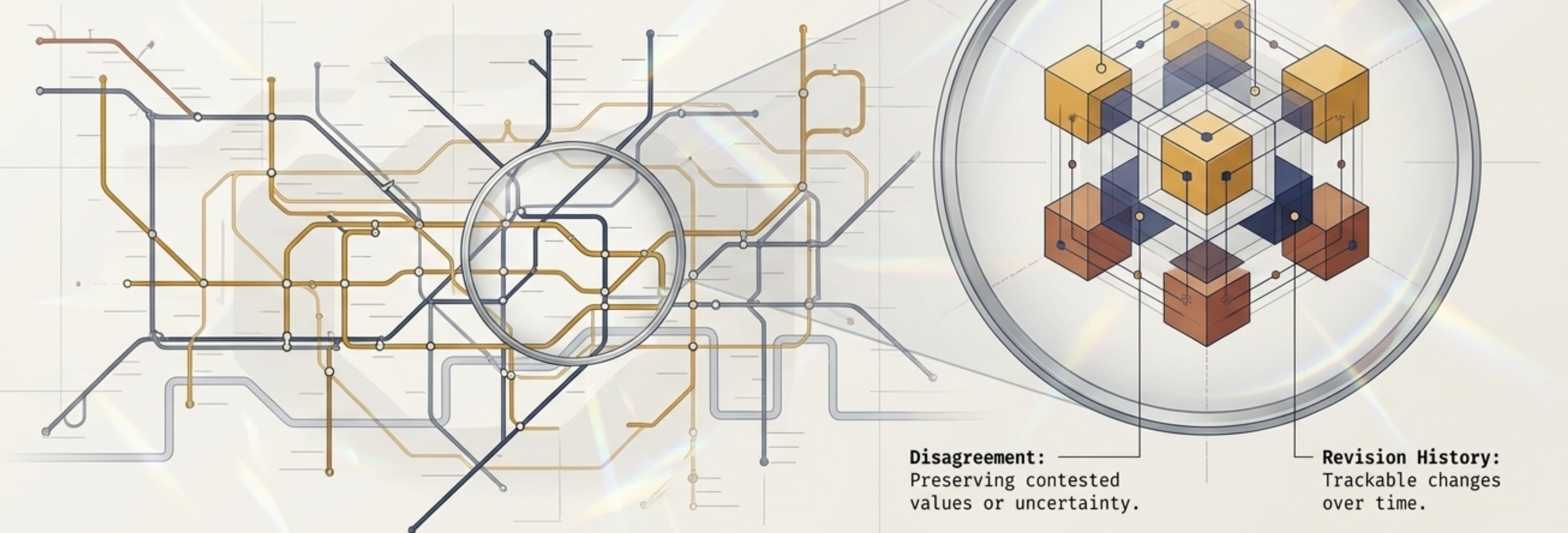


- Infrastructure enforces permissions: The surrounding execution system determines if an action is permitted, not the model's instructions.
- Independence is not automatic: Using the same model twice does not establish independent judgment. Where true independence matters, review may require additional evidence, different methods, or human oversight.



# The Verified Context Graph becomes the ultimate differentiator

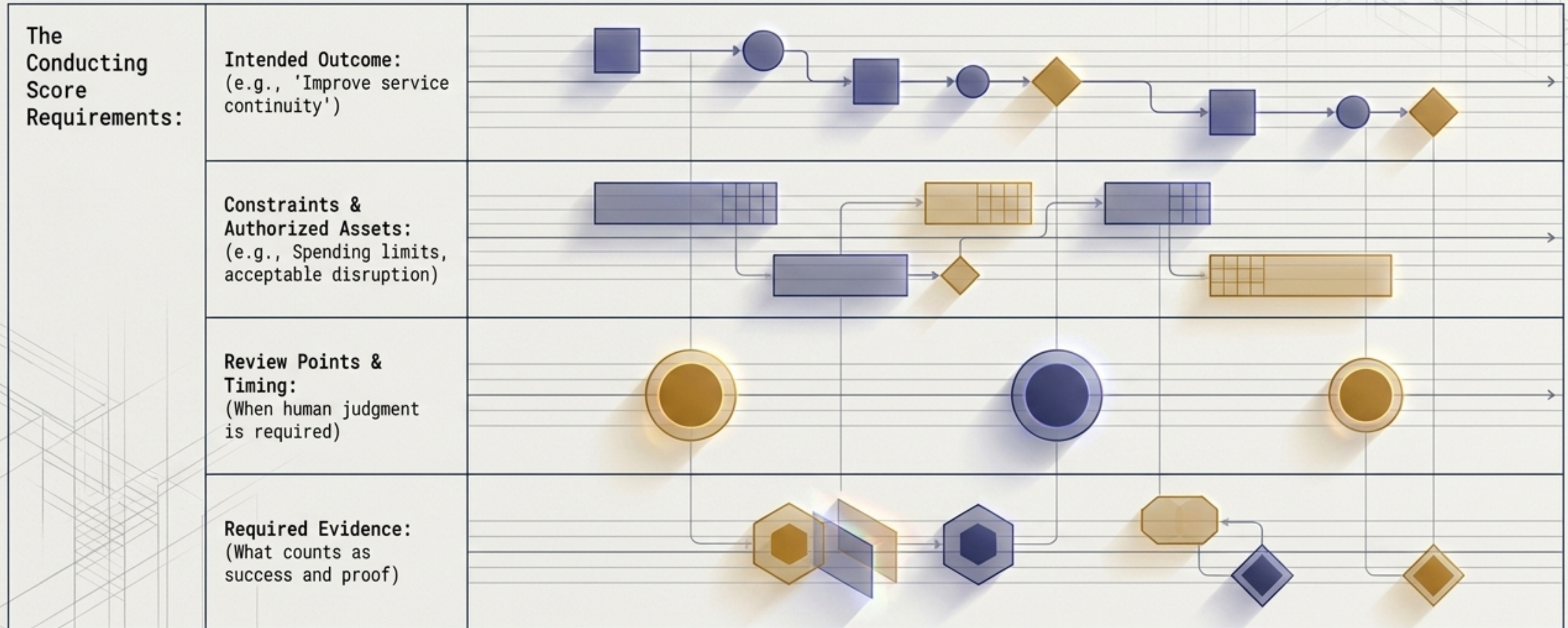
Domain knowledge becomes a maintained organizational resource. But verification must be interpreted carefully: an authenticated record can still contain an error.





# The shift in human work: Direction and Proof

Instead of designing a digital employee for every job title, humans shift to a form of conducting: holding direction and coordinating action without manually performing every cognitive step.





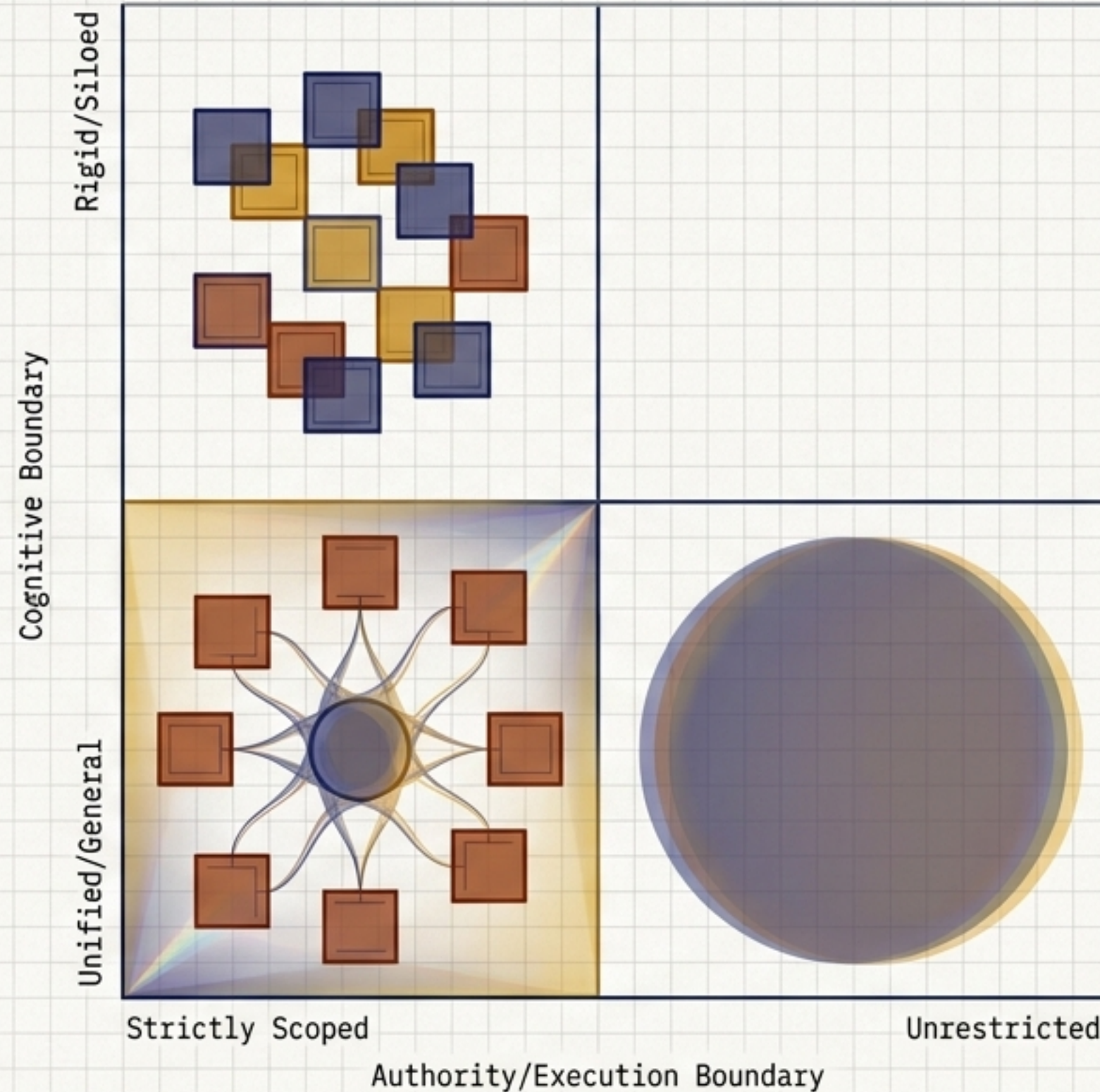
# Two ways to misbuild an AGI architecture

## The Rigid Network

Failure: Preserving unnecessary cognitive silos. Handoffs narrow context and duplicate interpretation.

## Authorised Reasoning, Bounded Action

Enables authorized reasoning across relevant domains while keeping consequential actions strictly bounded.

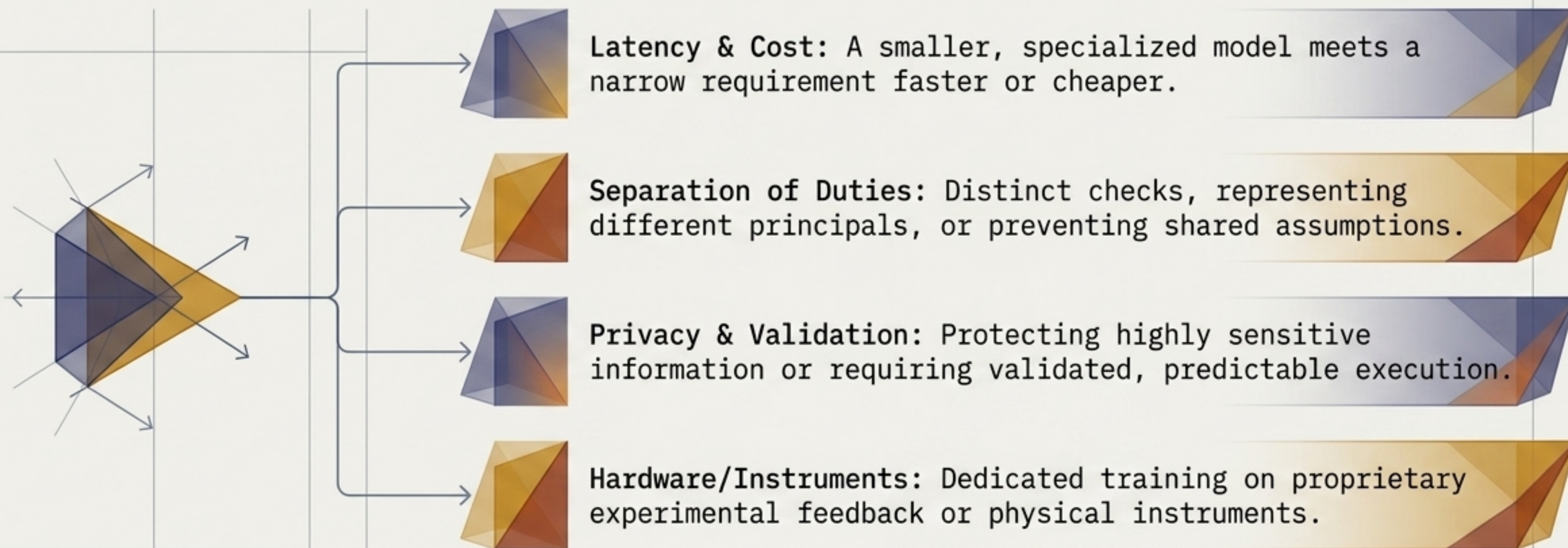


## The Erased Boundary

Failure: The "God-Agent". Erasing necessary boundaries allows a single mistake or compromise to affect the whole organization.



# When do specialist agents still make sense?

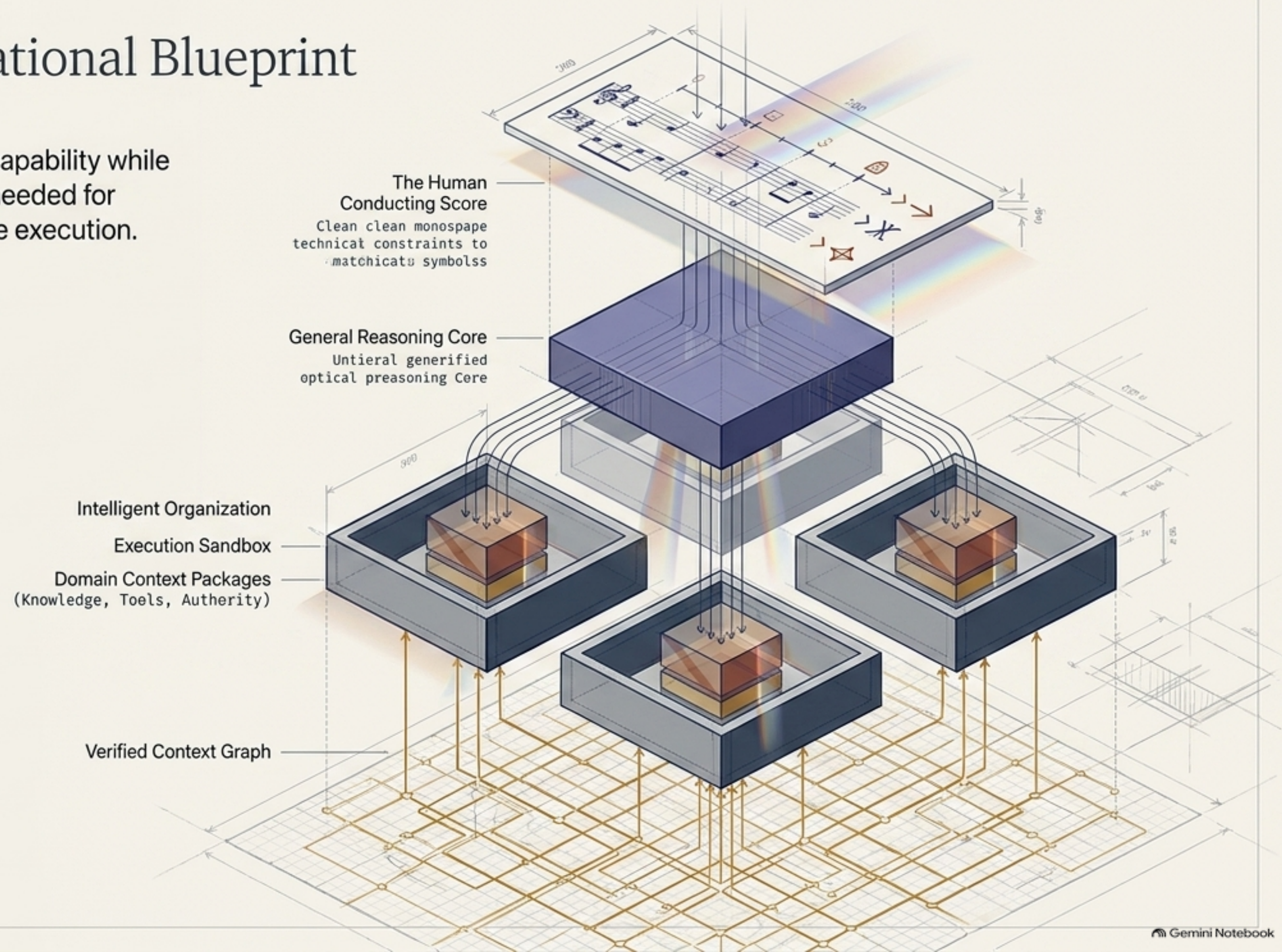


**The Testable Rule:** A domain label alone is no longer sufficient justification for a separate intelligence once general capability can reliably perform the work.



# The New Organizational Blueprint

Organizations can reuse general capability while maintaining the hard boundaries needed for privacy, accountability, and reliable execution.





**With AGI, domain-specific intelligence becomes optional.**

**Domain-specific context and accountable action remain essential.**